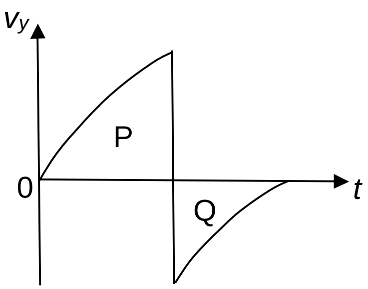
Examiner's Report with answers for Paper 3

1(a)	Displacement is defined as the <u>distance moved in a specific (or certain) direction</u>.	1
(b)(i)	$s_y = u_y t + \frac{1}{2} a_y t^2$ $1.96 = \frac{1}{2}(9.81) t^2$ $t = 0.632$ s $v t = 2.53$ $v = 2.53/0.632 = \underline{4.00 \text{ m s}^{-1}}$	M1 M1 A1
(ii)	$v_y^2 = u_y^2 + 2a_y s$ $= 2(9.81)(1.96)$ $v_y = \underline{6.20 \text{ m s}^{-1}}$ **OR** $v_y = u_y + a_y t$ $= (9.81)(0.632)$ $= \underline{6.20}\ \underline{\text{m s}^{-1}}$ **OR** $mgh = \frac{1}{2} m v_y^2$ $v_y^2 = 2(9.81)(1.96)$ $v_y = \underline{6.20 \text{ m s}^{-1}}$	M1 A1
(iii)	$\tan\theta = \dfrac{v_y}{v_x} = \dfrac{6.20}{4.00} = 1.55$ $\theta = \underline{57.1°}$	1
(iv)	<u>9.81 m s⁻², downwards</u>	1
(c)	 ➢ Correct shape {AB: curve of decreasing gradient, B: vertical or almost vertical line, BC: curve of decreasing gradient} ➢ Area P greater than area Q **AND** magnitude of v_y just before impact is greater than magnitude of v_y just after impact.	 1 1